

## Designing a New Heat Exchanger

Our first step was to find a material that met several criteria: inexpensive, durable, lightweight, and non-permeable so that oxygen could not interact with the water inside the exchanger. After much research, we decided to use the same aluminum material used in sealed food bags. This material was lightweight, impermeable, and can be bought in bulk. I tested the seal strength of a few unlabeled plastic juice containers made of this material by hooking them up to pressurized air to see at what pressure the aluminum would pop. This was necessary as the water running through our heat exchanger would be held around 10 psi, and the seal needed to withstand that pressure without breaking. The factory bags held up to around 40 psi of air, proving that there was an effective way to seal them to hold the required pressure.



**Figure 1: Testing Setup for Factory Sealed Food Containers**

The next problem was figuring out how to seal the aluminum sheets into a shape that we would later settle on. My plan was to test sealing methods and make pouches with similar dimensions to the factory juice pouches that I had already tested to see how much pressure my seals could withstand. The initial thought was to melt them together, but I feared that this process would be too expensive and would take too long to create the desired shape for our heat exchanger. My next, and most successful, idea was to use an ultrasonic sewing machine that was normally used for sealing plastic materials together. I ran several tests using this method. I tested different settings on the sewing machine, creating many pouches to be hooked up to the pressurized air and have their "pop pressure" measured. Based on this, I knew the most effective setting on the sewing machine for creating the best seal for our heat exchanger.



**Figure 2: Some Pouches Created to Test Bond Strength**



**Figure 3: Material Used for Creating New Pouches**

The last step was to create the actual design for the heat exchanger that we wanted to manufacture. I knew I had to maximize the surface area in contact with the surrounding environment while also making the heat exchanger capable of holding in the water's pressure. My partner and I created a fingerlike design with one inlet tube dispersing into multiple tubes running the length of the heat exchanger until they came back together into one outlet tube. We made a miniature model of the design and tested it to make sure it could withstand the pressure required. After testing the smaller design, a six-foot long prototype was created. This final prototype withstood the pressure that was required until it burst due to a sealing problem at one of the connection points.



**Figure 4: Fingerlike Design for Heat Exchanger to Maximize Surface Area**



**Figure 5: Full Heat Exchanger Prototype**

Overall, the process allowed me to get outside of my comfort zone in the design area and gave me a brief overview on how an idea can turn into a new product. The process tested my problem-solving capabilities and forced me to research and learn about innovative technologies that I had never investigated before.